

O-LEVELS CS Paper-2

Bubble Sort

Sorting: Simply arranging data in a particular order. This can be ascending or descending.

Example:

3	10	7	15	2	25
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— Ascending →

2	3	7	10	15	25
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— Descending →

25	15	10	7	3	2
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BUBBLE SORT: This is an Algorithm that repeatedly compares adjacent element (sath walay element) and swaps them if they are in wrong order. The largest element bubbles on Top.

Example —

10	5	25
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↙ ↓ ↓ ↘ → compare



Ascending



—

5	10	25
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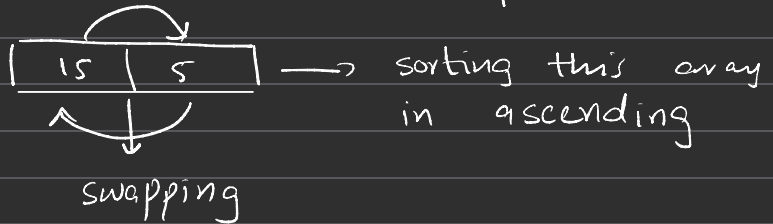
 → skip.

→ hence the array is completely sorted.

↳ if the previous element is smaller than the next element then swaps

Pass: A full traversal of the array during which adjacent elements are compared.
↳ From start to End.

Swap: Exchanging the values of two elements



5 | 15

Ascending Order: From smallest to largest
(e.g. 2, 4, 6, 8, 10)

Decending Order: From largest to smallest.

Q Why this sorting is called Bubble Sort?
This is called bubble sort because the largest value "Bubbles Up" to the top of the array in each pass, like a bubble rises in water.

PSEUDOCODE OF BUBBLE SORT

ADDITION

- FOR LOOP
- IF/ELSE
- Arrays.



100%
100%
100%

75%

50%

[10 | 1 | 3 | 15] → Ascending

[1 | 1 | 3 | 10 | 15] ←

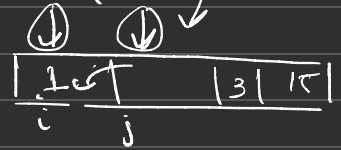
Temp = 10

100%

DECLARE array[1:4]: INTEGER
DECLARE i, j, temp: INTEGER

100%

FOR i ← 1 to 4
FOR j ← 2 to 3



IF array[i] > array[j]

v1 = 10

// swap

10 = 1

Temp ← array[i]

1 = v1

array[i] ← array[j]

array[j] ← Temp

END IF

NEXT j

NEXT i

END FOR